Critical Analysis of Nuclear Waste Disposal in Space

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Abstract:

Ever since the 1950s, humanity has dealt with the persistent problem of high-level nuclear waste (HLW). This material, which is a byproduct of civilian nuclear power and military defense programs, remains dangerously radioactive for tens of hundreds of thousands of years. While deep geological repositories are considered the most viable solution, they are faced with immense opposition and are a core reason for the opposition to nuclear power as a whole. Additionally, geological uncertainties make this storage method only a temporary solution. This has led scientists and engineers to revisit a radical and unconventional storage method, permanently removing this waste by launching it into space.

The concept of space disposal is not a new concept; in the 1970s and 1980s, NASA and other institutions undertook a systematic series of studies to assess its feasibility. With the increase in new aerospace technology, these solutions are more plausible. These analyses concluded that, from a technical and economic standpoint, the concept was sound. The physics of orbital mechanics and the economics of space launch, however complex, were surmountable. However, these series of studies consistently identified two critical barriers: the unacceptably high risk of catastrophic launch failure dispersing radioactive material into the Earth's

atmosphere, and the complicated restrictions of international treaties and principles that limit the use of nuclear material in space.

Most studies relating to the disposal of nuclear waste in space were completed pre-2010. This paper provides a critical re-evaluation of space disposal, moving beyond their historical test conclusions to analyze the results of historical conclusions and present-day realities. Specifically, it will explain current rocket advancements, such as reusable launch systems, and argue why innovations do not yet overcome fundamental safety concerns for nuclear waste transport. This paper will also project future advancements, exploring how emerging space technology could make this concept viable in the near future. Furthermore, this analysis will expand beyond the notion of placing waste into Low Earth Orbit, but detail specifically methodologies for planetary removal, including trajectories out of the solar system, or directly into the Sun. Covering the 1970s-1980s analysis and contrasting them with the technological landscape of today, this paper will conclude by outlining potential possibilities for implementation and assessing how near-future advancements could shift this proposal from a theoretical possibility to reality.

Introduction:

Current terrestrial storage has received significant opposition from both the government and the public. The Yucca Mountain repository in the United States serves as an example of the rejection surrounding terrestrial storage. Decades of research and billions of dollars were invested into this site, but the project was effectively terminated due to intense political opposition from the state of Nevada and concerns over geological stability, including the risk of water infiltration and seismic activity. A summary by the House Committee of Energy and Commerce states that “In 2010, despite Congress’s continued bipartisan support for Yucca

Mountain, the Obama Administration decided to terminate the project, attempting to withdraw the DOE's licence application from the NRC's review. The Secretary of Energy asserted that 'a geological repository at Yucca Mountain is not a workable option for long-term disposition' of nuclear waste, and DOE filed a motion with the NRC's Construction Authorization Board to withdraw the licensed application." This failure has left the U.S without a permanent disposal solution, forcing waste to be stored temporarily at reactor sites across the country. Similarly, other nations have encountered significant resistance. In Germany, the proposed Gorleben repository was abandoned after decades of public protest and scientific debate over the suitability of the salt dome geology. Even in countries making progress, such as Finland with its Onkalo repository and France with Cigeo, the process has been met with decades of delays, legal challenges, and public skepticism, highlighting the immense difficulty of achieving social acceptance with a terrestrial storage solution.

Given the significant issues with terrestrial disposal, there is a critical need to explore alternative pathways. By leveraging contemporary aerospace engineering and propulsion systems, there is potential to mitigate this burden. Integrating new technologies offers a theoretical pathway to remove existing waste from the earth entirely, therefore optimizing the energy production lifecycle. Recent breakthroughs in space technology have allowed for higher lift and carrying capacity of rockets such as Falcon Heavy.

A key advancement for this approach is the recent revolutionary invention in reusable rocket technology, which has drastically reduced the cost of access to space by utilizing recoverable boosters, such as SpaceX's Falcon 9, the Zhuque-3 (VTVL-1), and the Hyperbola-2Y. The cost of a launch can be reduced by up to 30% while also increasing the launch frequency and system reliability.

With the new changes in sociocultural, political, and environmental beliefs, this idea becomes a highly plausible solution towards the removal of nuclear waste on Earth. By using new rockets, which can carry approximately 63.8 metric tons to Low Earth Orbit, we can send multiple rockets and dispose of our nuclear waste.

In addition, this paper will also consider two disposal methods that were identified in the original studies as the only true long-term solutions:

1. Interstellar Ejection: The first and most permanent method is to use a series of gravitational assists to propel a payload containing nuclear waste out of the solar system entirely. This would place the material in a heliocentric orbit that never returns to the solar system, effectively removing it from the Sun's sphere of influence. This permanently eliminates any possibility of future interaction with Earth.
2. Heliocentric Injection: The second method is to use a direct, high-energy approach to de-orbit a payload into the Sun. This is a more challenging operation, as it requires canceling out the Earth's orbital velocity of approximately 30 km/s. However, the result would be total vaporization and atomization of the waste, removing the waste forever.

Having established the pathways for permanent waste removal, we can now address the legal and regulatory framework. The shift from a concept to reality introduces challenges that are not solved by engineering or law. The goal of intentionally launching high-level nuclear waste from the earth raises questions of international law, state responsibility, and the potential for catastrophic launch failure. The existing treaties, including the Outer Space Treaty of 1967

published by the UN General Assembly First Committee. The following articles are relevant to the problem at hand: Article I, Article II, Article IV, Article VI, Article VII, Article IX.

- Article I: States that the exploration and use of outer space shall be carried out for the benefit and in the interest of all countries.
- Article II: Prohibits national appropriation of outer space, including the Moon and other celestial bodies.
- Article IV: Prohibits placing in orbit around the Earth any objects carrying nuclear weapons or any other kinds of weapons of mass destruction, or installing such weapons on celestial bodies.
- Article VI: States that states parties shall bear international responsibility for national activities in outer space, whether carried on by government agencies or non-government entities, and for assuring that national activities are carried out in conformity with the provisions of the treaty.
- Article VII: Establishes that each state party that launches or procures the launching of an object into outer space is internationally liable for damage to another state party by such object or its parts.
- Article IX: Requires state parties to conduct all their activities in outer space with due regard to the corresponding interests of all other state parties, and to avoid harmful contamination and adverse changes in the environment of the Earth. It also states that if a state party has reason to believe that an activity or experiment planned by it or its nationals would cause potentially harmful interference with activities of other state parties, it shall undertake appropriate international consultations before proceeding.

Any proposal requesting the launch of nuclear waste into outer space must follow the Outer Space Treaty.

Method:

Our method has a very innovative and complex design, which will be broken down in our method. Our first step begins with us sending an electric spacecraft to low orbit to improve reusability and sustainability. We will send this autonomous spacecraft to the exosphere and will dock with the International Space Station (ISS). Second, the cargo is then loaded on the ISS, and the electric spacecraft will return to Earth to retrieve more cargo. In ISS, astronauts will load cargo into space capsules to send toward the sun, which is autonomous. Cargo gets destroyed in the sun's heat, and nuclear waste is obliterated.

I. Phase One: The Sustainable Earth-to-Orbit (ETO) Bridge:

The primary innovation here is the Electric Propulsion (EP) Shuttle. Using ion thrusters to power the spacecraft using reusable materials and resources. Unlike traditional chemical rockets, EP systems (like Hall effect thrusters or Ion drives) offer high specific impulse, meaning they use propellant much more efficiently over time.

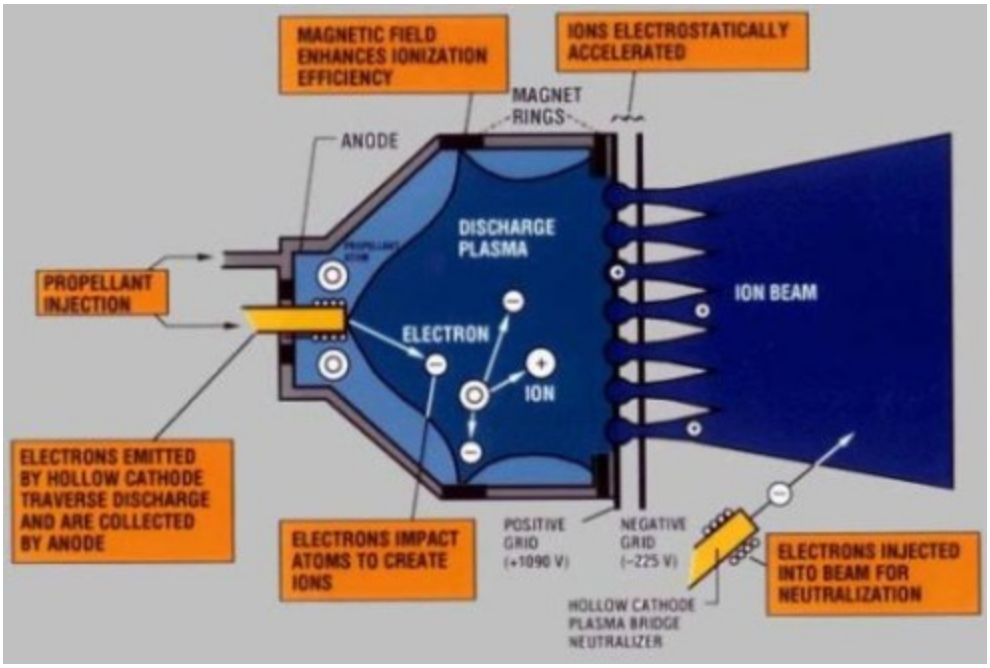
Propulsion Mechanics: Solar-Electric Ion Drives. To facilitate a sustainable "orbital ferry" between Earth and the International Space Station (ISS), our method utilizes Solar-Electric

Propulsion (SEP). Specifically, we employ high-efficiency Ion Drives using Xenon as a propellant.

1. The Ionization Process: The core of this system involves ionizing a noble gas, typically Xenon, by stripping away electrons to create a stream of positive ions. By utilizing solar energy collected via high-yield photovoltaic arrays, these ions are accelerated through an electrostatic field.

2. Momentum and Efficiency: Following Newton's Third Law of Motion, these ions are expelled from the thruster at exhaust velocities reaching approximately 90,000 m/s: Unlike chemical rockets that burn fuel quickly for high thrust, Ion Drives provide a much higher specific impulse. This allows the spacecraft to remain in the exosphere for extended periods without the need for frequent, heavy refueling. Sustainability: Because the energy source is solar and the propellant consumption is minimal, the "Earth-to-Orbit Bridge" becomes a low-cost, repeatable loop for waste transport.

3. Application to Nuclear Logistics: In the context of nuclear waste disposal, the Ion Drive's steady, reliable thrust is ideal for the precise maneuvering required to dock with the ISS. It provides the "long-duration" capability needed to move heavy shielded canisters safely through the exosphere without the volatile risks associated with traditional liquid-fuel combustion.



1. Low Earth Orbit (LEO) Logistics: The spacecraft operates as a reusable "tug." By staying within the exosphere and LEO, the craft avoids the massive structural stresses of atmospheric reentry, significantly extending its operational lifespan.

2. Energy Sourcing: To remain truly sustainable, these shuttles would utilize high-efficiency solar arrays to power their ion engines, creating a carbon-neutral transport loop between the Earth's upper atmosphere and the ISS.

From Earth orbit, the waste is sent out on capsules, reusing as much waste as possible to send out of the solar system or into the sun itself. Thereby completely removing it from low Earth orbit.

Navigation and Control (GNC):

For the mission to succeed without constant human intervention, the spacecraft utilizes a sophisticated Guidance, Navigation, and Control (GNC) system. This architecture ensures the safe transit of hazardous materials through the exosphere to the docking port.

1. Guidance: Path Planning

The guidance system calculates the optimal trajectory from the launch injection point.

Fuel Optimization: It uses "Lambert's Problem" algorithms to determine the most fuel-efficient transfer orbits, ensuring the Ion Drive's low-thrust profile is utilized effectively.

Collision Avoidance: The system cross-references real-time data from the Space Surveillance Network (SSN) to automatically adjust its path and avoid orbital debris.

2. Navigation: State Estimation

The navigation sub-system determines the spacecraft's current position and velocity (its "state").

Sensors: It employs a suite of Star Trackers, Inertial Measurement Units (IMUs), and Global Navigation Satellite System (GNSS) receivers.

Relative Navigation: As the craft nears the ISS, it switches to LiDAR (Light Detection and Ranging) and optical cameras to measure the exact distance and orientation (pose) relative to the station's docking adapter.

3. Control: Precise Maneuvering

The control system executes the commands from the guidance system by translating them into thruster firings.

Reaction Control System (RCS): While the Ion Drive provides the primary forward thrust, a secondary set of cold-gas or electric micro-thrusters manages the "attitude" (the direction the craft is pointing).

Six Degrees of Freedom (6DoF): The GNC system manages movement across all axes, allowing for the "soft-docking" required to safely transfer nuclear cargo without damaging the ISS structure.

II. Phase 2: Orbital Cargo Integration:

Once the autonomous electric shuttle arrives, the mission shifts from transport to payload processing. This phase is designed to ensure maximum safety before the final "no-return" trajectory toward the Sun is initiated.

1. Automated docking procedure: The system uses GNC (Navigation and Control) to dock.
    a. Robot hand-off: Robotic hand-off for less risk. The station’s robotic arm (like the Canadarm2) extracts the shielded nuclear canisters from the shuttle’s cargo bay.



b. Radiation Containment: The cargo remains in Universal Transport Casks (UTCs) containers made from multi-layered containers made of lead, boron-carbide (for neutron absorption), and a titanium outer shell to prevent any leakage onto the station.

2. Assembly Solar Insertion Capsules (SICs): The OIF serves as the point of assembly for the Solar Insertion Capsules (SICs). While the shuttles are built for reuse, the SICs are "single-use" kinetic projectiles designed for one-way travel.
    a. Ballistic Securing: Specialized station robots secure the waste canisters in the SICs.
    b. Ablative Shielding: Each SIC is equipped with a thick carbon-carbon heat shield. This isn't for atmospheric reentry, but for the intense thermal flux. It represents heat flow in a specific direction (vector quantity), commonly driven by temperature differences from hot to cold. Types include conduction, convection, and radiation, essential for optimizing thermal insulation, electronics cooling, and energy efficiency. The heat flux equation provides a comprehensive, vector-based picture of how heat moves—both its magnitude and direction—rather than just a single temperature reading.
        i. The heat flux equation is:

$$q'' = -\frac{k \cdot \Delta T}{\Delta x}$$

3. Safety Verification: The OIF is an area where robots can perform final checks toward safety.
    a. Leak Detection: Sensors on the OIF scan the canisters for gamma-ray spikes or structural micro-fractures caused by the launch.
    b. Trajectory Calibration: The OIF mission control calculates the precise "departure window." Because the Sun is a moving target and Earth is orbiting at high speeds, the SIC must be released at the exact millisecond to ensure it doesn't accidentally enter a stable orbit or fly toward another planet.

III. Phase 3: Terminal Obliteration:

Once the Solar Insertion Capsule (SIC) is verified and cleared for departure from the ISS, it goes through the most energy-intensive part of the mission. Our main goal is for the cargo to crash into the sun and not go “toward it”.

1. The Physics of Solar Entry: Earth orbits the Sun at approximately 30(km/s) (67,000 mph). To hit the Sun directly from Earth's orbit, a spacecraft must fire its engines in the opposite direction of Earth's travel to cancel out that speed.
    a. Problem Statement: Canceling 30 (km/s) requires more fuel than almost any modern rocket can carry.
    b. The Innovative Solution: Instead of a direct "brute force" burn, your method utilizes a Jupiter or Venus Gravity Assist. To optimize fuel efficiency and reduce transit time, the capsule utilizes a Gravity Assist (or 'Slingshot') maneuver. By

performing a close flyby of a celestial body—such as Venus or Mercury—the spacecraft exchanges momentum with the planet's gravitational field. This maneuver effectively modifies the capsule's heliocentric velocity without the need for extensive propellant consumption. This strategic use of orbital mechanics allows the SIC to achieve the high-energy trajectory required for solar insertion while staying within the strict mass-to-fuel ratios of the electric transport model.

Analysis:

The system itself is deemed to be theoretically possible, but the bridging between theory and practice is huge. A critical analysis is required to determine if true waste disposal is the plan in reality, not just on paper.

Looking at the Launch Capacity, the Falcon Heavy represents the current peak of operational launch capacity, with a payload capacity of 63.98 metric tons to Low Earth Orbit. With a global inventory of approximately 450,000 metric tons of high-level nuclear waste, simple division reveals a requirement of over 7050 successful Falcon Heavy launches just to transport existing waste to LEO. This number increases substantially when considering the additional mass of containment vessels, which typically add 50-100% more mass for radiation shielding and impact protection. When LEO is reached, there are still future steps. There is often a misconception with space disposal where the amount of energy required to reach disposal trajectories is underestimated. Reaching LEO requires approximately 9.4 km/s of delta -v. While achieving true disposal requires more. A solar escape trajectory, where waste is ejected from the solar system, requires an additional 12-16 km/s delta-v from LEO, typically achieved through

gravity assist from Jupiter or other planets. Direct injection into the sun requires canceling the Earth's 30 km.s orbital velocity, requiring approximately 24-30km/s delta-v from LEO.

Pro:

Waste takes resources and pollutes the Earth, making it detrimental for future generations.

Counterargument (from dad):

There is already a lot of nuclear waste from Voyager 1 and 2, plus. And lots of waste on the moon. If an accident occurs, more waste is thrown into orbit. Even if this works, humans will want to send more waste, causing more pollution in space, along with not knowing how the sun will react. Will this be cost-efficient? Closed plans for maybe reusing waste.

Links for works cited

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